🔹 Introduction

Open-source software forms the backbone of modern computing systems. From smartphones to servers, most systems rely on software developed collaboratively and shared freely. This project analyzes the Linux Kernel, one of the most important open-source projects, focusing on its origin, philosophy, Linux implementation, ecosystem, and comparison with proprietary alternatives.

🅰️ PART A — ORIGIN & PHILOSOPHY

A1. Origin Story

The Linux Kernel was created by Linus Torvalds in 1991 when he was a student. At that time, Unix systems were expensive and restricted. Linus wanted a free and flexible operating system for personal use. Instead of keeping it private, he released it publicly, allowing others to contribute.

This decision solved:

- High cost of proprietary OS
- Lack of customization
- Restricted access to source code

Linux quickly evolved into a global collaborative project.

A2. License Analysis (GPL v2)

Four Freedoms:

1. Freedom to run software
2. Freedom to study source code
3. Freedom to modify
4. Freedom to distribute copies

✔ Linux Kernel provides all four freedoms.

Can companies modify & sell?
Yes, but they must release modified source code (GPL rule).

GPL vs MIT:

- GPL → strict (must share changes)
- MIT → flexible (can keep changes private)

My choice: GPL (ensures openness)

Free beer vs freedom:

- Free beer = no cost
- Freedom = control over software
  Linux = freedom, not just free.

A3. Ethics of Open Source

Should all software be open source?
✔ Pros:

- Transparency
- Innovation
- Security

❌ Cons:

- Business limitations
- Security risks if misused

Is profit on OSS ethical?
Yes, if companies contribute back (e.g., Red Hat).

Standing on shoulders of giants:
Means building on existing knowledge.
Open source accelerates innovation.

🅱️ PART B — LINUX FOOTPRINT

Installation:

- Pre-installed in most distros
- Managed via package managers

Directories:

- Kernel: `/boot`
- Config: `/etc`
- Logs: `/var/log`

User & Group:

- Runs as root (important for system control)

Service Management:

systemctl status
systemctl start
systemctl stop

Updates:

- Via apt/yum updates
- Community patches

🅲 PART C — FOSS ECOSYSTEM

Dependencies:

- GCC compiler
- GNU tools
- Libraries

Inspired tools:

- Android OS
- Ubuntu, Fedora

LAMP relation:

- Linux is “L” in LAMP stack

Community:

- Developers worldwide
- Maintained by Linus Torvalds

🅳 PART D — COMPARISON

Dimension	Linux Kernel	Windows Kernel
Cost	Free	Paid
Security	Open audit	Closed
Support	Community	Official
Modification	Allowed	Not allowed
Control	Community	Corporate

Verdict:
Linux is better for flexibility and learning, while Windows is easier for general users.

🧾 Conclusion

Linux Kernel demonstrates the power of collaboration and openness. It is reliable, secure, and widely used. I would recommend it for real-world deployment and would contribute to it in the future.

💻 3. ALL 5 SHELL SCRIPTS (READY)

✅ Script 1

```
#!/bin/bash
# System Identity Report

STUDENT_NAME="Your Name"
SOFTWARE="Linux Kernel"

KERNEL=$(uname -r)
USER=$(whoami)
```

```
UPTIME=$(uptime -p)
DATE=$(date)
DISTRO=$(cat /etc/os-release | grep PRETTY_NAME | cut -d '"' -f2)

echo "===== SYSTEM REPORT ====="
echo "Name: $STUDENT_NAME"
echo "Software: $SOFTWARE"
echo "Kernel: $KERNEL"
echo "User: $USER"
echo "Uptime: $UPTIME"
echo "Date: $DATE"
echo "Distro: $DISTRO"
echo "License: GPL (Open Source)"
```

✅ Script 2

```
#!/bin/bash
# Package Inspector

PACKAGE="linux"

if dpkg -l | grep -q $PACKAGE; then
  echo "$PACKAGE installed"
  dpkg -l | grep $PACKAGE
else
  echo "$PACKAGE NOT installed"
fi

case $PACKAGE in
  linux) echo "Linux Kernel: backbone of open source OS" ;;
  apache2) echo "Apache: web server" ;;
  mysql) echo "MySQL: database system" ;;
  firefox) echo "Firefox: open browser" ;;
esac
```

✅ Script 3

```
#!/bin/bash
# Disk Auditor

DIRS=("/etc" "/var/log" "/home")
```

```
for DIR in "${DIRS[@]}"
do
  if [ -d "$DIR" ]; then
     PERM=$(ls -ld $DIR | awk '{print $1,$3}')
     SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
     echo "$DIR => $PERM | $SIZE"
  fi
done
```

✅ Script 4

```
#!/bin/bash
# Log Analyzer

LOGFILE=$1
KEYWORD=${2:-error}
COUNT=0

if [ ! -f "$LOGFILE" ]; then
  echo "File not found"
  exit 1
fi

while read LINE
do
  if echo "$LINE" | grep -iq "$KEYWORD"; then
     COUNT=$((COUNT+1))
  fi
done < $LOGFILE

echo "$KEYWORD found $COUNT times"
```

✅ Script 5

```
#!/bin/bash
# Manifesto Generator

read -p "Tool: " TOOL
read -p "Freedom means: " FREEDOM
read -p "Build: " BUILD

FILE="manifesto.txt"
```

```
echo "On $(date), I believe in open source." > $FILE
echo "I use $TOOL daily." >> $FILE
echo "Freedom means $FREEDOM." >> $FILE
echo "I will build $BUILD and share it." >> $FILE

cat $FILE
```

📘 4. README.md (COPY THIS)

```
# OSS Audit Project

Name: Your Name
Roll No: XXXX
Software: Linux Kernel

## Scripts

1. system_report.sh
2. package_inspector.sh
3. disk_auditor.sh
4. log_analyzer.sh
5. manifesto.sh

## How to Run

chmod +x script.sh
./script.sh

## Requirements
Linux system (Ubuntu recommended)
```